



In Class assignment 1

Researching Django in the Real World

Objective

The goal of this assignment is to understand **why Django is used in professional software development**, what types of applications can be built with it, and how companies use Django in real-world products.

Django is a Python-based web framework designed for rapid development, security, scalability, and building a wide range of web applications. The official Django website lists organizations such as Instagram, Mozilla, Pinterest, Disqus, and National Geographic among sites using Django.

Instructions

1. Research each question using the **internet and reliable sources**.
2. Answer **all 10 questions**.
3. Write the answers **in your own words**.
4. **Do not copy and paste** answers directly from websites, ChatGPT, or other sources.

Research Questions

1. **Why should a developer choose Django?**
Research the main advantages of Django compared with building a web application from scratch.
2. **What problems does Django solve for web developers?**
Investigate features such as authentication, database handling, URL routing, forms, security, and the built-in admin panel.

3. What types of applications are best suited for Django?

Research whether Django is better suited for websites, business applications, APIs, e-commerce, social networks, dashboards, or data-driven applications.

4. What are the advantages and disadvantages of Django?

Identify at least **5 advantages** and **3 disadvantages** of using Django in a real-world project.

5. Which major companies use Django?

Research companies such as **Instagram, Pinterest, Mozilla, Disqus, Bitbucket, and The Washington Post**, and determine how Django is/was used in their technology stacks.

6. What real-world products have been built using Django?

Find at least **5 real products or platforms** built with Django and explain what each product does. Examples include social platforms, content platforms, commenting systems, and developer tools.

7. Can Django handle millions of users?

Research how Django applications scale and investigate examples of high-traffic websites that have used Django. Explain what additional technologies—such as caching, databases, load balancing, background workers, or APIs—may be required.

8. Why is Django popular for data-driven applications?

Research Django's ORM, database support, admin interface, and Python ecosystem. Explain why these features can be useful for applications that work with large amounts of structured data.

9. Django vs Flask vs Node.js: Which should a company choose?

Compare Django with Flask and Node.js/Express based on **development speed, scalability, security, database support, learning curve, ecosystem, and typical use cases**.

10. Is Django still relevant for modern software development?

Research Django's current adoption, community, job market, recent releases, API development, cloud deployment, AI integration, and real-world use in 2026. Then give your own conclusion: **Would you choose Django for a new project today, and why?**